浙江大学 2005－2006 学年秋冬季学期

《　　普通物理 II　　》课程期末考试试卷

开课学院：　理学院　，考试形式：闭卷，允许带　__计算器_入场

考试时间：　_2006_年__01__月_ _13___ 日，　所需时间：　120　分钟

考生姓名：　_____ ____ _ _____学号：　_____专业：　_____ _____

题序	一	二(1)	二（2）	二（3）	二（4）	二（5）	二（6）	总　分
得分								
评卷人								

Ⅰ. Fill in the space underlined. (50%)

1. Figure 1 shows a Thomson atom model of helium (He, **Z=2**). Two electrons, at rest, are embedded inside a uniform sphere of positive charge **2e**. The distance d of _____between the electrons is so that the configuration is in static equilibrium.

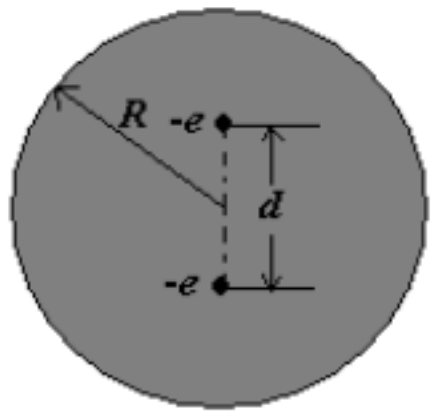


Figure 1

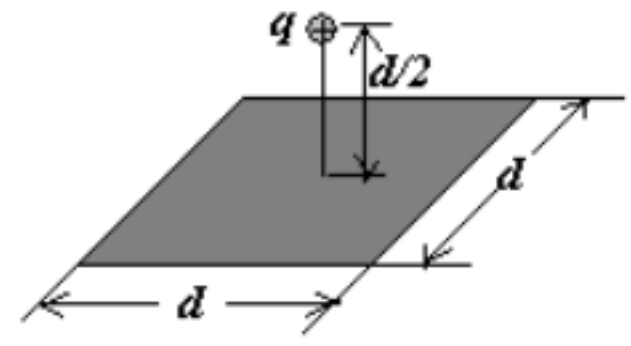


Figure 2

2. A point charge **+q** is a distance **d/2** from a square surface of side **d** and is directly above the center of the square as shown in Fig. 2. The electric flux through the square is of_____.

3. A resistor is in the shape of a truncated right circular cone (Fig.3). The end radii are **a** and **b,** and the length is **L**. If the tape is small, we may assume that the current density is uniform across any cross section. The resistance of this subject is_____.

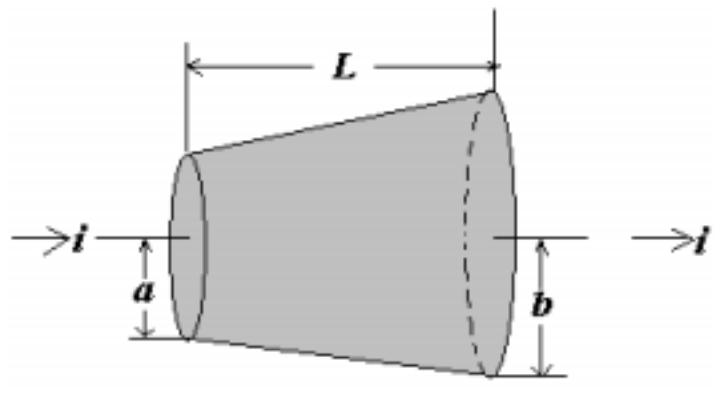


Figure 3

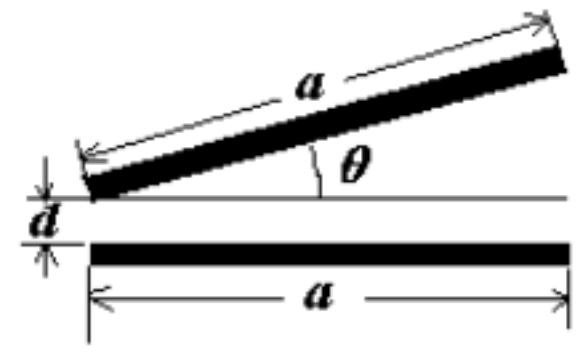


Figure 4

4. A capacitor has square plates, each of side **a**, making an angle θ with each other as shown in Figure 4. The capacitance is given by_____ for small θ.

5. As shown in Figure 5, a current i_1 flows in a circular loop of radius **R**. An infinite long wire carrying a current i_2 is in the plane of the loop, and at a distance **d>R** from the center of the loop to the wire. The force between the wire and the loop is_____.

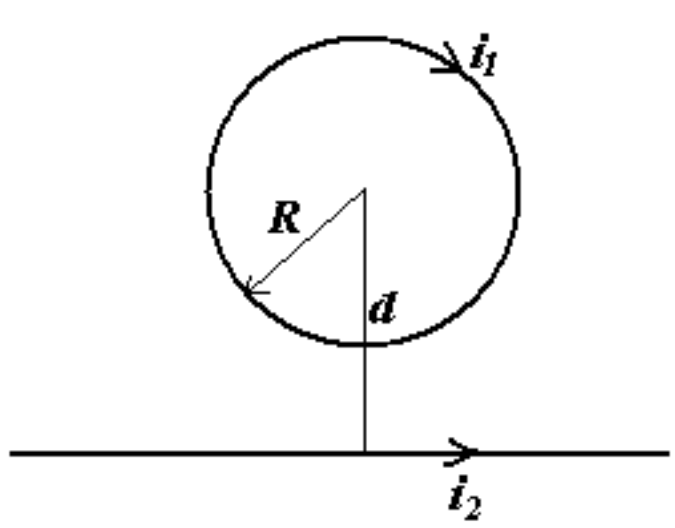


Figure 5

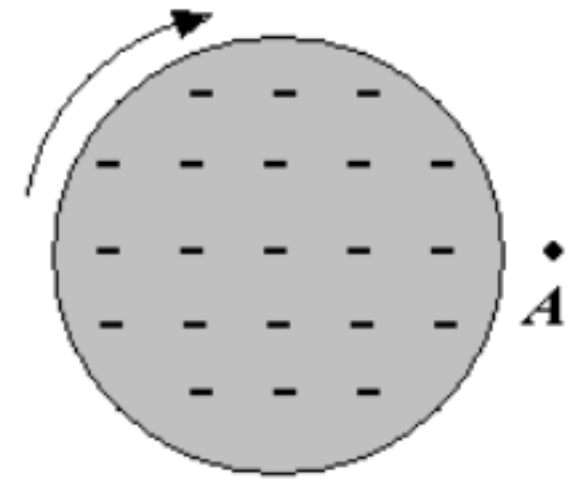


Figure 6

6. The negatively charged disk in Fig. 6 is rotated clockwise. The direction of the magnetic field at point **A** in the plane of the disk is _____.

7. A rectangular loop of wire with length **a**, width **b**, and resistance **R** is placed near an infinitely long wire carrying current **i**, as shown in Fig 7. The distance from the long wire to the loop is **D**. The current in the loop is _____ as it moves away from the long wire with speed **v**.

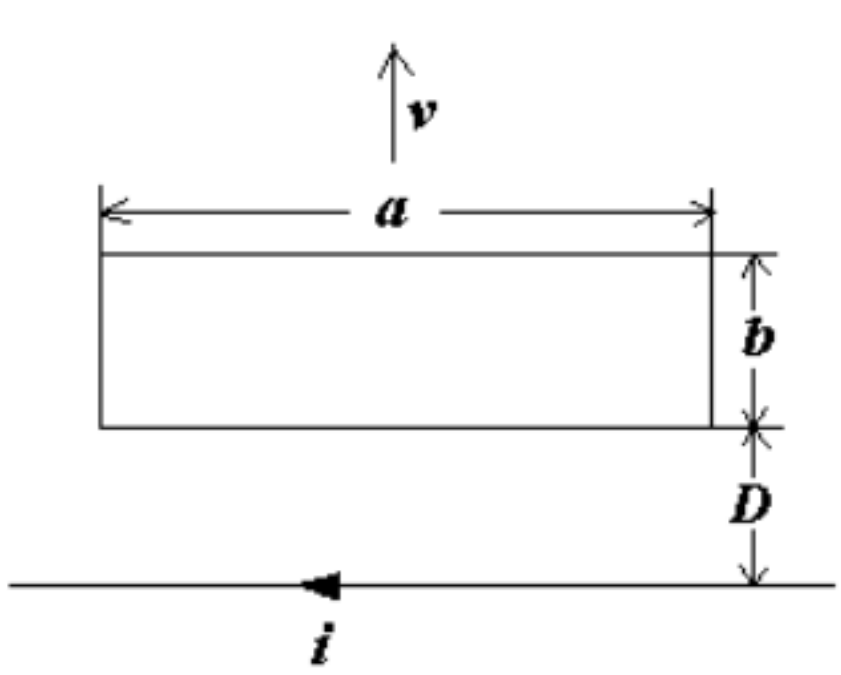


Figure 7

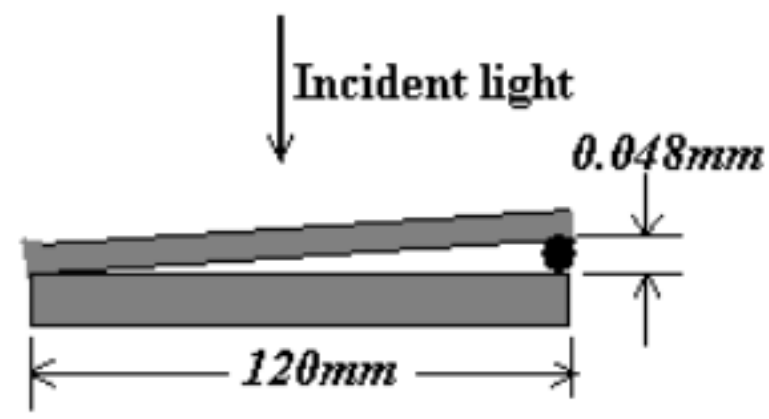


Figure 8

8. A broad source of light (λ**=680nm**) illuminates normally two glass plates **120mm** long that touch at one end and are separated by a wire **0.048mm** in diameter at the other end (Fig. 8). How many bright fringes appear over the **120-mm** distance? _____.

9. The wavelength of a **1.00-keV** electron is of _____ .

10. Two ideal polarizing sheets are stacked so that none of the incident unpolarized light is transmitted. A third polarizing sheet is slipped between the first two sheets at an angle of **45**^{o} to the bottom sheet. The fraction of light transmitted through the entire stack is _____.

II. Problems (present the necessary equations in solution) (50%)

1. (8%)Please write out:
   (a) The image formation equation for the spherical refracting surface with paraxial rays.
   (b) The properties of electromagnetic wave at distance from a wave source.

2. (7%)What would be the action of a **quarter-wave plate**($\lambda/4$ 波片)on
   (a) Linearly polarized (assume the plane of vibration to be at 45^0 to the optic axis of the plate), and
   (b) Circularly polarized light.

3. (10%)A grating (**N=5000**) is illuminated by two monochromic lights with the wave length of **600** and **400 nm** respectively. The **m** th principal maximum of the former light is meet the **m+1** th principal maximum of the later at **3 cm** from the central fringe on the screen. The focus length of the lens is **50 cm**. Find the grating constant **d**, and the typical width of the principle fringes.

4. (13%)Figure 9 shows a parallel-plate capacitor being charged. The voltage across the plates has a time dependence $V=V_0\cos\omega t$. The fringing of the electric field and radiation can be ignored.
   a) To determine the electric and magnetic field in the region between plates as a function of time **t**.
   b) How large does the current flow in the wires and what is the time dependence of the current density between the plates?
   c) Write the Poynting vector **S** at the edge of the capacitor (including its magnitude and direction).

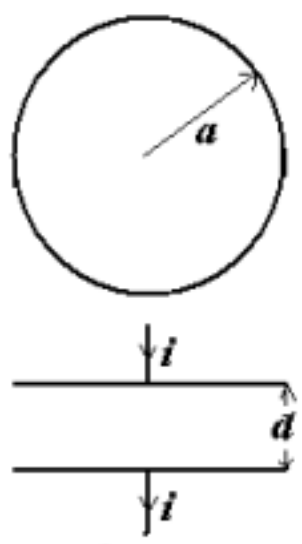


Figure 9

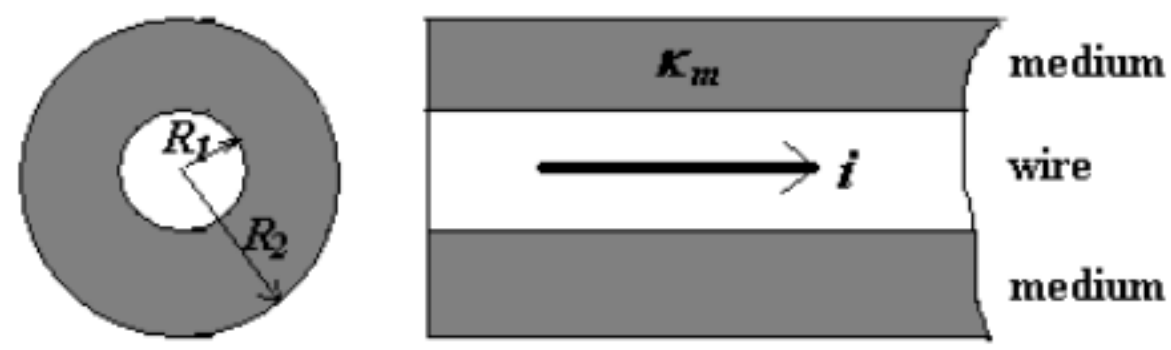


Figure 10

5. (12%)As shown in Figure 10, a long cylindrical wire of radius R_1 carries a current **i**, distributed uniformed over the cross section of the wire. And it is surrounded with a magnetic medium (磁性介质) of permeability constant κ_m, and with outer radius R_2.
   (a) Calculate the magnetic field strength **H**, the magnetic induction strength **B** in and out of the magnetic medium.
   (b) The induced current **i** (束缚电流) per unit length on the inner and outer surface of the magnetic medium.